

Implementation Summary

This file records what was implemented from the two-agent plan and what remains as the next mathematical work.

1 New Artifacts

1. `level-set-deficit-identity.md`

This implements the Nicola–Tilli route up to an exact theorem:

$$\frac{2}{|\Omega|} (E(\Omega) - E(B)) = \frac{1}{|\Omega|} \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2 \omega_n^{2/n} m(t)^{1-2/n}} dt.$$

It also proves the weighted variance identity

$$\int_{\Sigma_t} \frac{(|\nabla u| - \bar{f})^2}{|\nabla u|} = \frac{m(t)}{P(t)^2} D_H(t).$$

This gives a precise bridge from the convexity gap in `faber-krahn-1.pdf` to approximate Serrin data on good superlevel sets.

2. `quantitative-selection-principle.md`

This implements the BDV selection route as a single-set lemma. Given

$$\varepsilon = \alpha(\Omega_0), \quad \delta = E(\Omega_0) - E(B_1), \quad q = \delta/\varepsilon \ll 1,$$

it selects a minimizer of

$$E(U) + f_{\hat{\eta}}(|U|) + \sqrt{\varepsilon^2 + \tau^2(\alpha(U) - \varepsilon)^2}, \quad \tau = q^{1/4},$$

and proves the quantitative preservation estimates

$$\begin{aligned} \alpha(\tilde{\Omega}) &\geq (1 - Cq^{1/4})\alpha(\Omega_0), \\ E(\tilde{\Omega}) - E(B_1) &\leq C(E(\Omega_0) - E(B_1)). \end{aligned}$$

Hence

$$Q_\alpha(\tilde{\Omega}) \leq CQ_\alpha(\Omega_0).$$

2 What Is Now Proved or Reduced

The Talenti/Nicola–Tilli route now has a clean first theorem: the global Saint-Venant deficit is exactly the integrated level-set loss. This is a standalone quantitative refinement of the proof in `faber-krahn-1.pdf`.

The selection-principle route now has a concrete replacement for the sequence argument in BDV Proposition 4.4: the selected set preserves the relevant deficit-to- α ratio up to constants.

The compactness bottleneck has been localized. The remaining non-explicit part is not the penalized minimization; it is the quantitative entry into the global nearly spherical graph regime.

3 Next Targets

1. Write the finite-perimeter version of the level-set identity.

Replace smooth regular-level arguments by a.e. coarea statements for the torsion function. This should be mostly technical and self-contained.

2. Prove a boundary-layer propagation lemma.

The target form is

$$\exists t \downarrow 0 \quad \text{with} \quad |\Omega \setminus \{u > t\}| \text{ small} \quad \text{and} \quad D_H(t) + D_I(t) \text{ controlled.}$$

This is the main missing step for making the level-set route prove sharp Fraenkel stability of the original domain.

3. Track the Alt-Caffarelli flatness constants for selected minimizers.

The needed statement is

$$\alpha(U_*) \leq \varepsilon_0(N, R, \tau) \Rightarrow \partial U_* \text{ is a global } C^{1,\gamma} \text{ graph over } \partial B_1.$$

The new note already proves the preceding Hausdorff-closeness estimate from density and α .

4. Collect stable Serrin inputs compatible with the level-set variance.

Existing stable Serrin theorems often use oscillation or Lipschitz norms of the normal derivative. The level-set route naturally gives a weighted L^2 variance of $|\nabla u|$, so the next search should focus on Serrin stability results accepting integral boundary data.